Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/17/2012
PI Name:	Karma, Alain Ph.D.		
Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a.karma@neu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617 373 2929
Organization Name:	Northeastern University		
PI Address 1:	Department of Physics		
PI Address 2:	360 Huntington Ave		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	11/16/2010	**End Date:**	11/15/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Trivedi, Rohit (Iowa State University)		
Grant/Contract No.:	NNX11AC09G		
Performance Goal No.:			
Performance Goal Text:			

Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime.
While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites

Task Description:	from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. This aim will be pursued by developing a new coarse grained dendritic network approach that is capable of simulating quantitatively the long-range diffusive interactions between primary, secondary and higher order branches of typical hierarchical dendritic tree structures on experimentally relevant length and time scales. Validation of this approach will be carried out through quantitative comparisons with phase-field simulations as well as both microgravity and ground based experiments in well-characterized aluminum based binary alloys, such as Al-Cu and Al-Si. Those studies will set the stage for planning future microgravity Bridgman experiments on the International Space Station that use bi-crystalline samples to investigate both intragrain and intergrain dendrite interactions in a diffusive growth regime.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	We developed a novel multi-scale modeling approach in order to investigate quantitatively the solidification of large dendritic arrays on a grain scale much larger than the diffusion length. The “Dendritic Needle Network” (DNN) model is designed to predict individual dendrite dynamics at the scale of the coarse grain, in the concentrated alloy limit where dendrites form highly ramified tree-like structures with thin needle-like primary and higher order branches. This approach represents the dendritic network as an ensemble of thin needles and tracks the network evolution on the grain scale by formulating two conditions at two distinct length scales that rigorously prescribe the growth rate of each needle in the limit where the tip radius is much smaller than the solute diffusion length. The unique scale-bridging advantage of this approach is that it describes the evolution of the entire dendritic network and interactions of branches on the grain scale without tracking the solid-liquid interface dynamics on the tip scale, as in a phase-field simulation. The model, implemented on massively parallel Graphic Processing Units (GPU), is capable of simulating quantitatively the interactions between individual branches and grains in spatially extended and highly branched dendritic structures, in the early transient regime as well as in the steady state. It permits to investigate fundamental aspects of microstructure evolution, such as the dendritic spacing selection, at the scale of the experiment. We validated quantitatively the DNN simulations against analytical and numerical solutions in the case of isothermal solidification of an equiaxed crystal. Then, we extended the model to directional solidification conditions (i.e. imposed temperature gradient and growth velocity). Hence, we may apply the model to the directional solidification of an Al-7wt.%Si alloy, used in experiments performed onboard the International Space Station in the framework of the CETSOL project, and produce quantitative comparisons with both microgravity and ground based experiments. As a first study, we identified the limits of stability of primary spacing for given processing conditions (namely, a pulling velocity V=10 µm/s and a temperature gradient G=100 K/cm). For these parameters, within an error margin of ˜ 20 % on the steady tip supersaturation compared to phase-field simulations, the DNN model gives results that would require a mesh size about 20 times smaller with phase-field, i.e. simulations about 25000 times longer. Finally, for the exact conditions of a microgravity experiments, i.e. V=10 µm /s and G=10 K/cm, the DNN model predicts a dendritic spacing that agrees well with the observed microstructures. The very next application of the DNN model will be the achievement of multi-grain simulations with interaction of misoriented grains in directional solidification, which will be benchmarked with phase field simulations and directly compared with microstructures solidified in microgravity. The next foreseen modeling challenge is the extension of the approach to three dimensions, which will require a new mathematical formulation of the solute flux condition at the tip. Additionally, we extended a quantitative phase-field (PF) formulation for directional to polycrystalline growth. Significant implementation effort was dedicated to improving the computational efficiency of the existing phase-field model, and hence to broaden the range of applicable time and length scales. The equations were reformulated using a non-linear preconditioning of the order parameter, which enables to use a grid size – and consequently a time step – much higher than the usual values taken so far. Furthermore, the model has been implemented for massively parallel computing on GPUs, reaching an additional speed-up of about 20 to 30 times. The resulting computationally efficient PF model may now be applied to simulate domains extended over several millimeters to centimeters during experimentally relevant time scales, such as minutes to hours. For the well-documented Succinonitrile-Acetone alloy, we achieved an extensive computational study of the tertiary branching mechanisms and shaping of the grain boundaries during the directional solidification of a two-grains microstructure in two dimensions. We explored a wide range of experimental parameters, namely the pulling velocity, V = 15 to 35 µm/s, the temperature gradient, G = 10 to 50 K/cm, and the grain misorientation angle, a = 5 to 45°, in order to determine the influence of these fundamental parameters on polycrystalline growth. A significant amount of simulation results has thus been gathered and still needs to be thoroughly processed and analyzed in the very next future, in order to draw conclusions on the critical mechanisms of diffusive grain interactions and the dynamical shaping of grain boundaries. Furthermore, the study was so far limited to a configuration with one grain oriented along the temperature gradient and pulling velocity direction while the other grain was misoriented. The study of the growth and interactions of two misoriented grains is part of the next foreseen investigations. Ultimately, the combined study of polycrystalline directional solidification with the DNN model together with phase-field simulations should provide quantitative comparisons with both microgravity and ground based experiments. On the one hand, this study will help understanding long-standing fundamental questions such as the history-dependent selection of spacing or the non-monotonic nature of the spacing-velocity relationship. It will open the way to the study of dynamical intragrain and intergrain selection of dendritic spacing and the formation of grain boundaries, determining for the final properties of industrial processed material. On the other hand, the quantitative and reliable modeling of directional solidification experiments at a realistic scale will provide a precious tool for planning future microgravity Bridgman experiments on the International Space Station that use bi-crystalline samples to investigate both intragrain

	and intergrain dendrite interactions in a diffusive growth regime.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Abstracts for Journals and Proceedings	Tourret D, Karma A. "Novel needle-network multi-scale model for the solidification of highly branched dendritic microstructures." TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. TMS 2012 Meeting 141st Annual Meeting, Orlando, FL, March 11-15, 2012. Conference Program, p. 110. http://www.tms.org/Meetings/Annual-12/PDFs/TMS2012FinalProgramWEBwithAbstracts.pdf , Mar-2012
Articles in Peer-reviewed Journals	Tourret D, Karma A. "Multi-scale needle-network model of complex dendritic microstructure formation." IOP Conference Series: Materials Science and Engineering. 2012;33:012095. MCWASP XIII: 13th International Conference on Modeling of Casting, Welding and Advanced Solidification Processes, Schladming, Austria, June 2012. http://dx.doi.org/10.1088/1757-899X/33/1/012095 , Jun-2012